

BOOKS Brief it like a new Master's student

- B Bounds** — what must it not do?
never modify raw snapshots; Python + numpy/matplotlib; ask before steps > 10 min
 - O Objective** — what should it achieve?
measure how the 10/50/90% Lagrangian radii evolve; find core collapse
 - O Output** — what comes back?
lagrange.py + one labelled figure + 5-line note
 - K Knowns** — what must it know?
snapshots in run01/, N-body (Hénon) units, scales in the header
 - S Success** — how will we judge it's good?
radii at t=0 match the IC file within 1%; report Myr and pc
- Objective = what to achieve · Output = what you get back · Success = how we judge it**

GUARD Second round — when the first answer isn't good enough

- G Grounding / tools** — point it at the real sources and tools (ADS, your files, your code) and ask it to cite where each claim comes from
- U Uncertainty** — allow “I don't know”; ask it to flag what it is unsure about
- A Audience / role** — who the result is for, and at what level
- R Reference examples** — if it keeps getting it wrong, show it one good example (an iteration step, not a first-round must)
- D Done checks** — say how it must check its own work before calling it done (tests, a sanity plot, a known limit)

Verify Let the agent act on reversible steps; keep a human gate on irreversible or science-conclusion steps.

Checks: **git diff** · **tests** · **independent reference calculation** · **convergence check** · **known limits** · **physical sanity**

Resources

code.claude.com/docs/en/best-practices
learn.chatgpt.com/docs/codex/cli
Going further: github.com/mattpocock/skills
docs.nersc.gov/development/coding-agents
github.com/adsabs/scix-mcp
Basic AI Skills (Moodle): moodle.uni-heidelberg.de/course/view.php?id=27651

Data note

Heidelberg University guidance: don't enter sensitive, personal or confidential data, or copyrighted text, into AI systems that process data outside university infrastructure.

Your turn

1 One task you repeat in your research:

2 Split it into 3–6 steps; mark each: agent / human check

Step	Describe the step	agent	human check
1		☐	☐
2		☐	☐
3		☐	☐
4		☐	☐
5		☐	☐
6		☐	☐

For one agent step: Success — how I’ll judge it: _____

Output — what comes back: _____

3 Now brief it with BOOKS

B

 Bounds

O

 Objective

O

 Output

K

 Knowns

S

 Success